

Airframe Technology Master

Master Class

Lecture 2

Best and Final Offer:

A Cost Engineering Stand Point

- Understand customer value
- What is Cost Engineering?
- Life Cycle Cost
- Procurement importance and its process
- Contract types
- Cost estimate : Airframe Cost drivers
- Cost to price factors
- Gap analysis between offer and should price
- **Best And Final Offer**

- All these aircraft have something in common.....



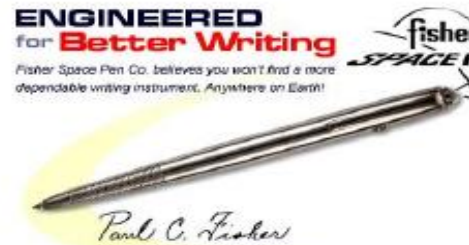
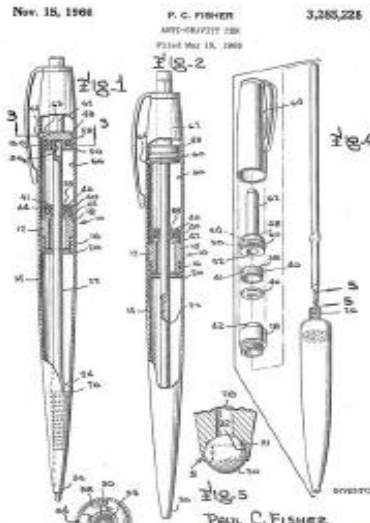
Understand Customer Value



Need to write in low gravity environment ?

Apolo

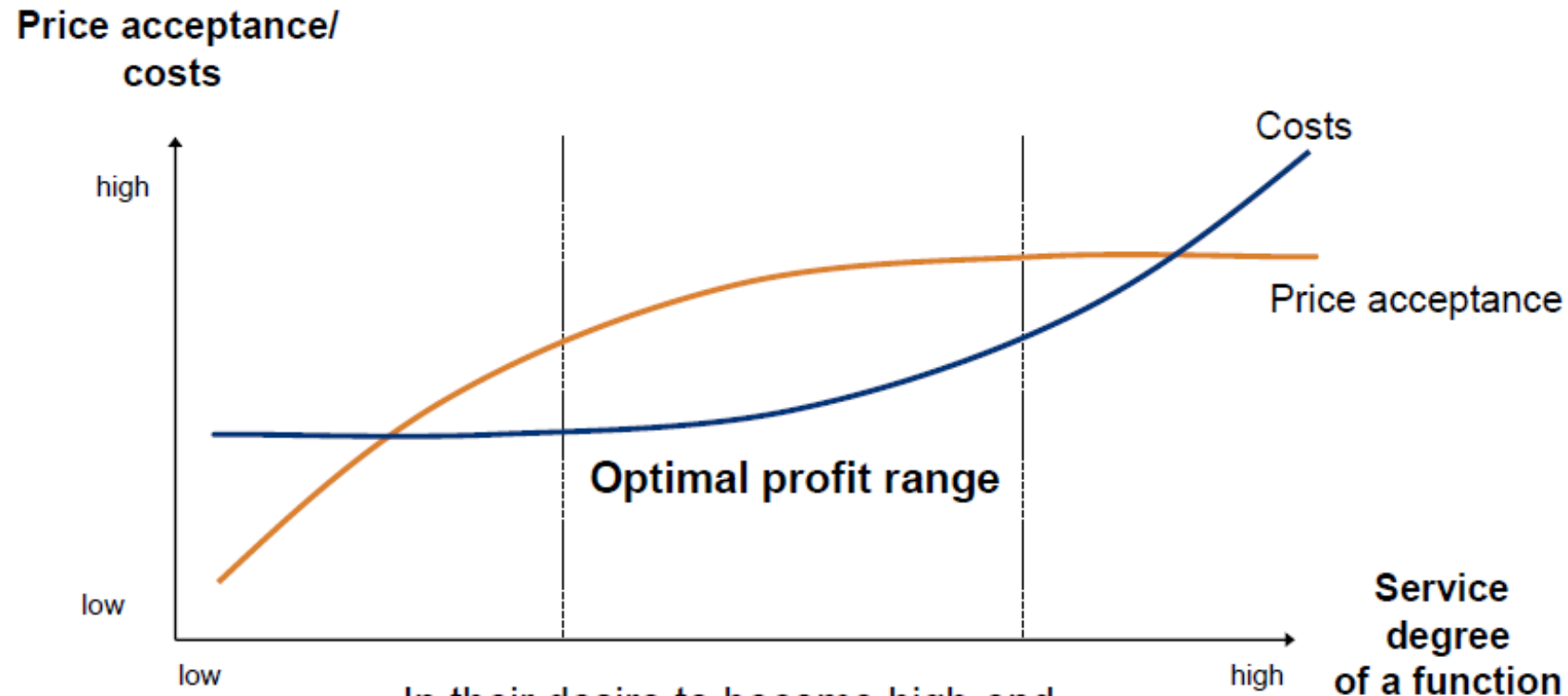
Soyuz



In 1965 a private company spent **several M\$** developing a ballpen

<https://www.spacepen.com/>

One requirement can lead to several answers
Adding cost to the decision process allow to identify the real need



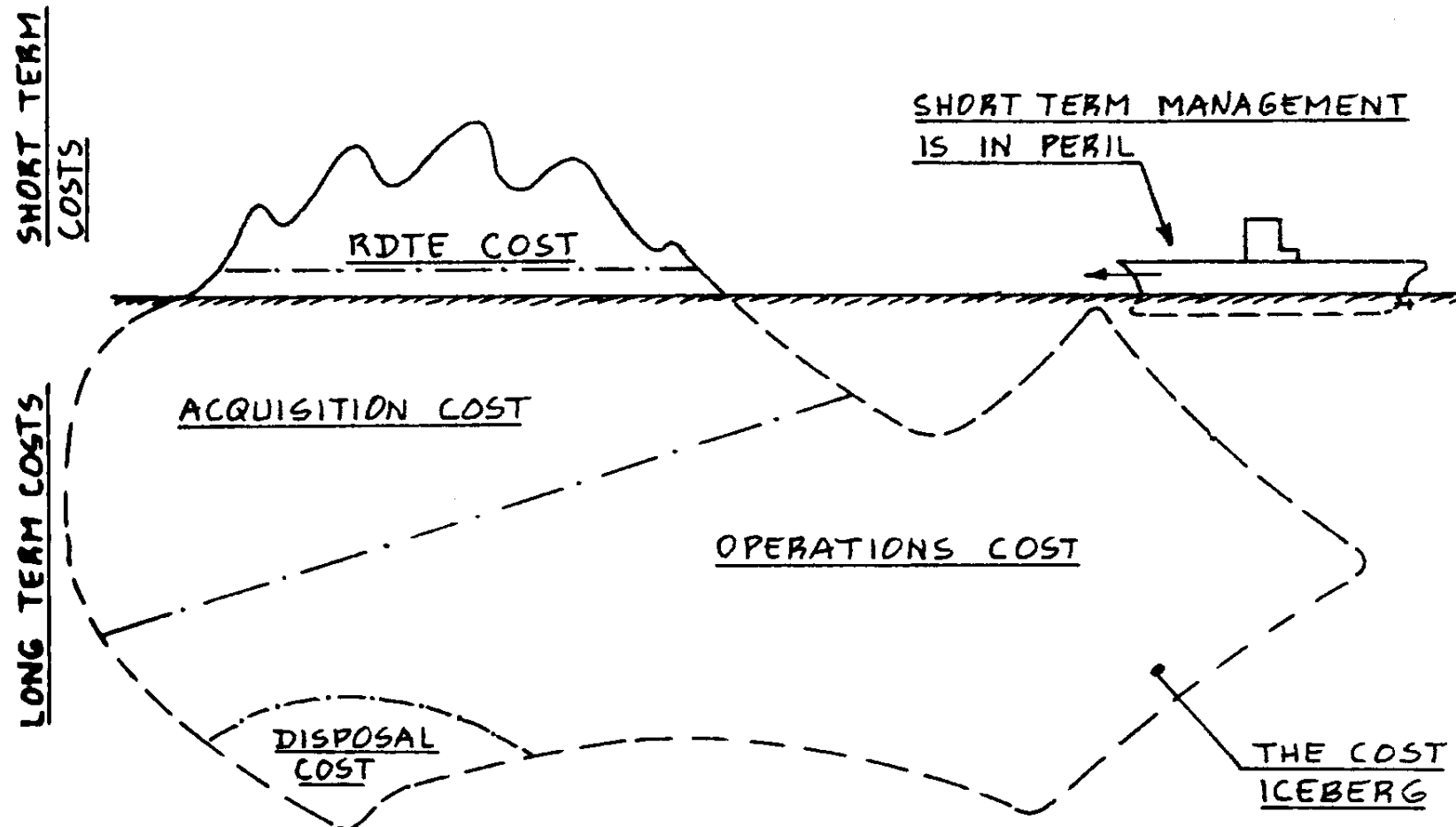
In their desire to become high-end manufacturers, many firms leave their optimal profit range

Cost Engineering is the application of **engineering technical knowledge and expertise on technologies and products** to deliver **cost information** to internal customers for better cost sensitive decision making

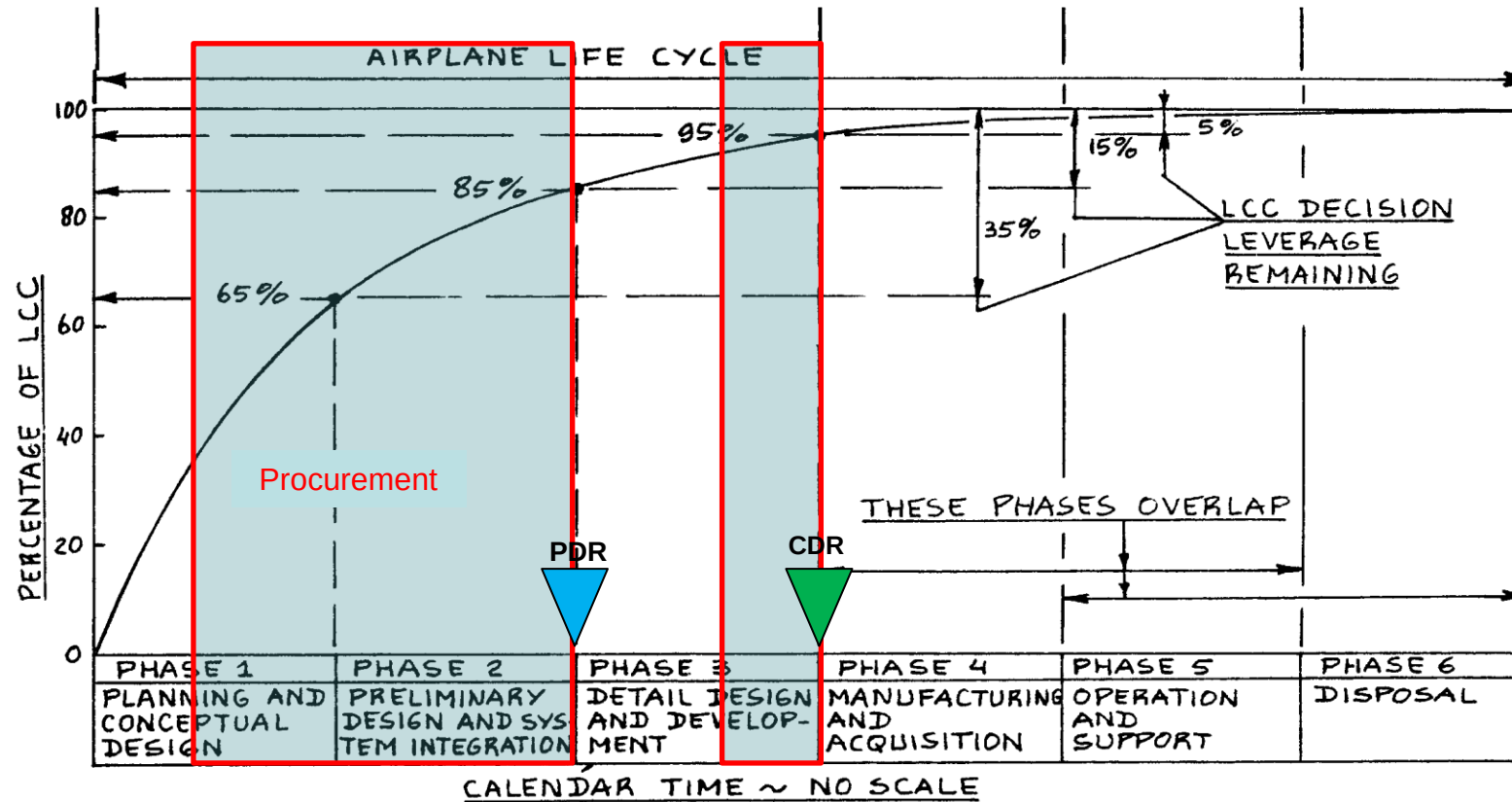
- Cost engineering **provides cost estimates and highlights cost drivers**, allowing key operational functions to prioritize their efforts,
- Cost engineering delivers an additional cost view, complementary from operational functions traditional bottom up .
 - ✓ Provide **independent, reliable, transparent** Cost estimations,
 - ✓ Provide quick & reliable initial **cost baseline**,
 - ✓ Identify **cost reduction opportunities & synergies** ,
 - ✓ Provide **risk and oportunities** cost assessment,
 - ✓ **Cost structure** standardization for better cost control.

Key benefits:

- ✓ Provide to Program & Procurement an additional leverage for setting **value-for-money** target costs / target prices with internal and external Suppliers,
- ✓ Fostering a **cost reduction mentality** by improving the understanding of the correlation between technical drivers and cost,
- ✓ Establish **standardized** deliverables & processes across the product life cycle,
- ✓ Application of a **common set of cost estimation methods & tools**.

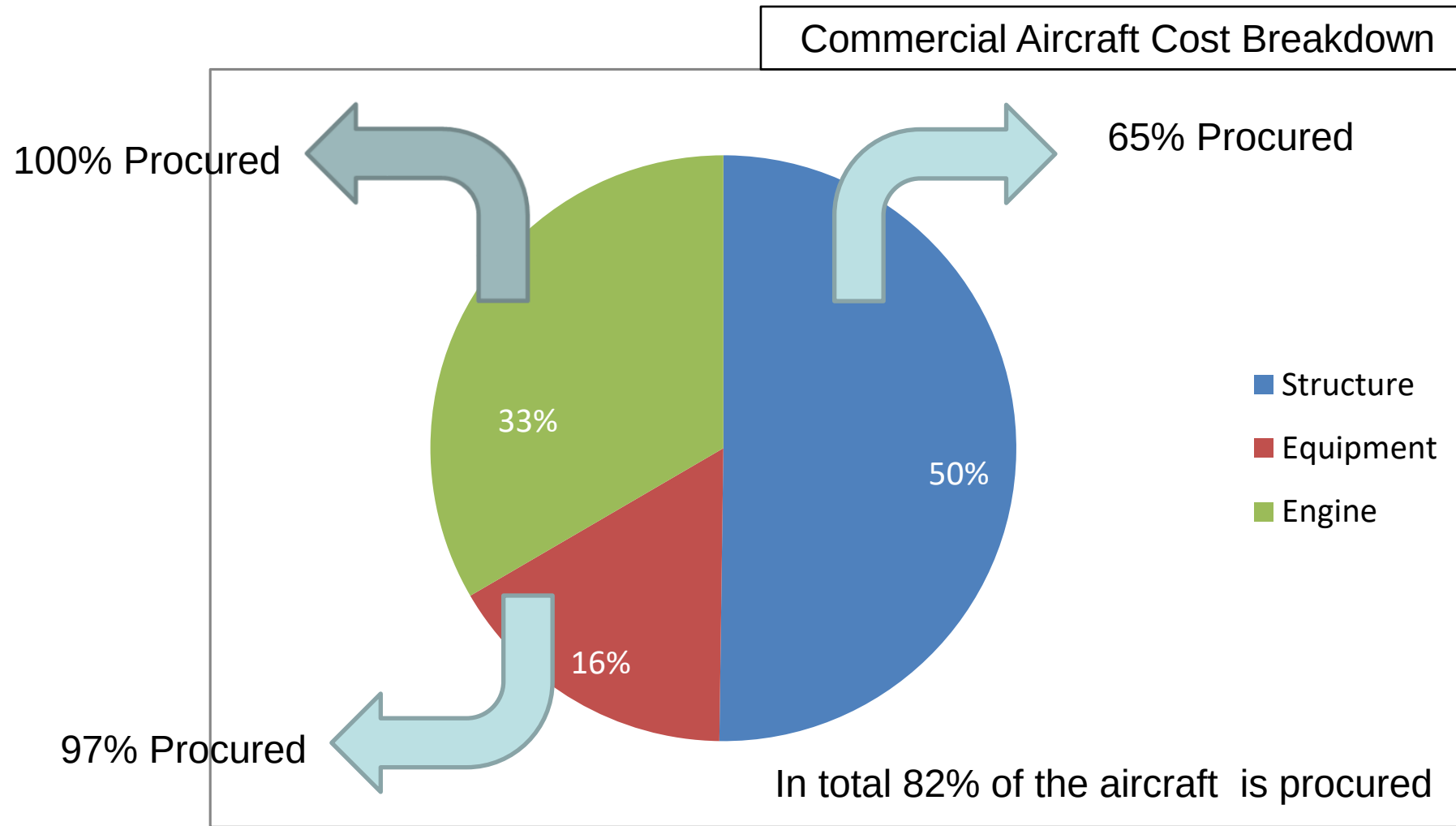


Procurement Importance



Depending on the kind of contract (see later) the information available at the moment of the offer could be very limited/in-mature.

Procurement Importance



Procurement Process

- Make or Buy analysis
 - Technology / Facilities
 - Strategy
 - Cost
- Initial Market analysis (RFI – Request For Information)
 - Request to a wide selection of capable suppliers.
 - The answer is a **price indication** i.e.: 1,5 M€ - 1,7 M€
 - This process helps to:
 - Select a reduced number of capable suppliers
 - Confirm/challenge the internal “should price” target
- Request For Quotation
 - Request to the reduced number of suppliers selected in the RFI for a formal commercial offer
 - The answer is a **price** i.e. : 1,65 M€

Procurement Process (cont)

- Supplier selection Process
 - Select the supplier based on:
 - Capabilities
 - Reliability
 - Price
 - Company Strategy
- Negotiation
 - Analysis of the gap between the supplier price and the internal should price
 - Agree on the terms and conditions (industrial and legal)
 - Request to the supplier their **BEST AND FINAL OFFER (BAFO)**

and.....



CONTRACT AWARD

By Scope

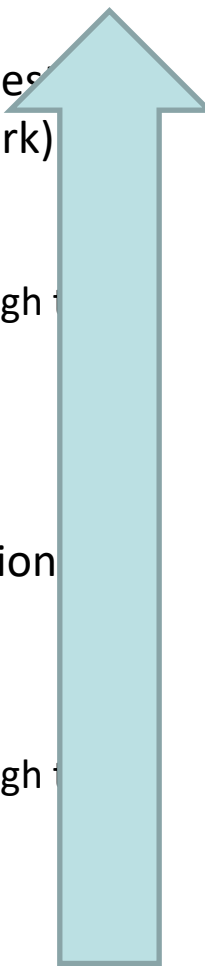
— Design and Build: supplier is responsible to design, manufacture and test the work package following the purchaser specification. (Statement of Work)

- Risk sharing
→ Supplier funds the complete activity and recover the investment through sales.
- No risk sharing
→ Buyer pays the development following and agreed milestone plan.

— Build to print: buyer delivers to the supplier the technical documentation (definition dossier).

- Risk sharing
→ Supplier funds the complete activity and recover the investment through sales.
- No risk sharing
→ Buyer pays the jigs and tools following and agreed milestone plan.

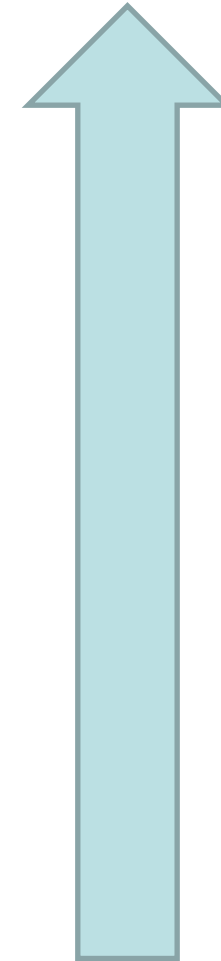
- Buyer supplies the jigs and tools



Supplier Risk

By Price Scheme

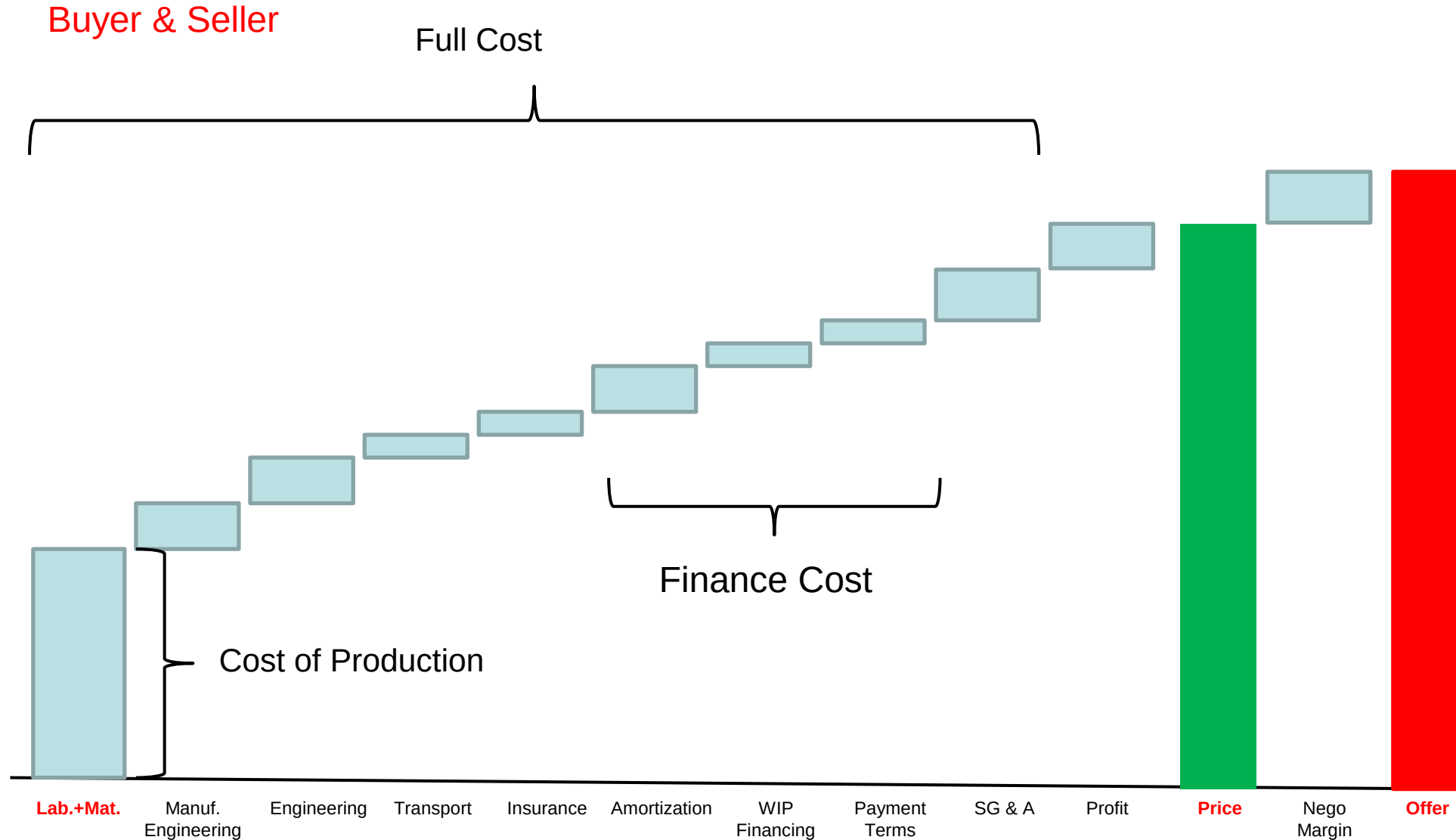
- Fix and firm : the price is the same all along the contract duration.
- Price evolution: the contract includes a clause to revise the price by
 - Batch
 - DateThe contract fix the price review formula or the prices per batch.
- Cost Plus: the price is based on supplier actual cost
- Time and material



Supplier Risk

Buyer & Seller

- Definition dossier maturity
- Program / project conditions (rate, year, contract type....)
- Market references (average hourly rate, low cost countries,...)
- Industrial set up
- Manufacturing process maturity.
 - Specific processes / machinery
 - Tolerances
 - Material
 - Exotic materials



Buyer & Seller

Once the suppliers short list is defined, this process starts

**Stage 1: Technical gap analysis**

*Gap between technical, industrial
and program assumptions*

Has the potential
to close ~ 50%
of the cases at WP level

Limited Cost Eng. activity



If gap not solved

**Stage 2: Economical gap analysis**

*At global WP level, or sub-WP level
Global cost and cost to price comparison*

Has the potential
to close ~ 40%
of the cases at WP level

Few weeks activity



If gap not solved

**Stage 3: Deep dive analysis**

*Very detailed approach including manuf.
process analysis, industrial footprint,
Engineering performance ...*

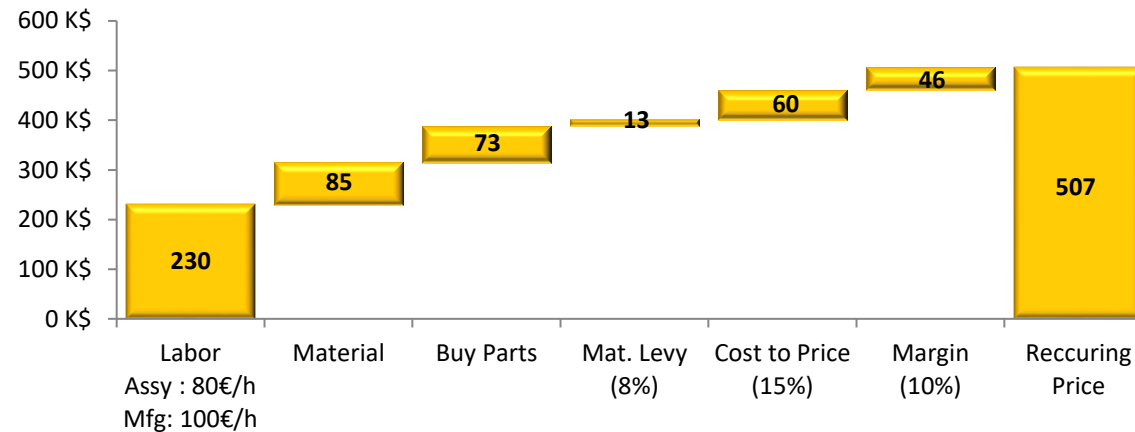
Has the potential
to close ~ 10%
of the cases at WP level

Some months activity

Should Price vs. Offer : example

Buyer & Seller

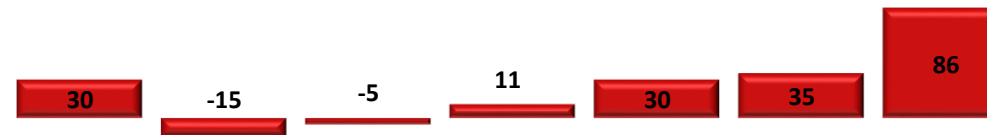
Supplier
„As-Is“
simulation



Assumptions

- 60 % in WE
- 40 % in East. Europe

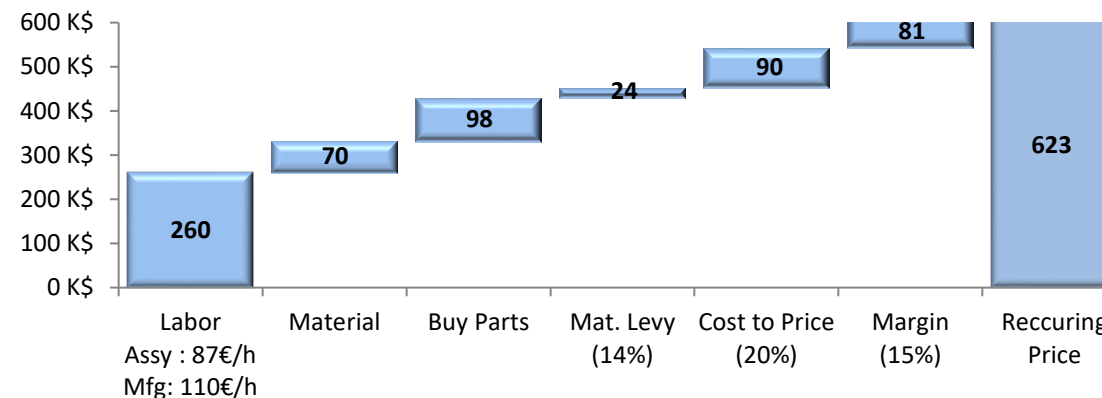
Gap : 14%



Gap Drivers

- Labor
- Cost to Price
- Margin

Supplier Offer

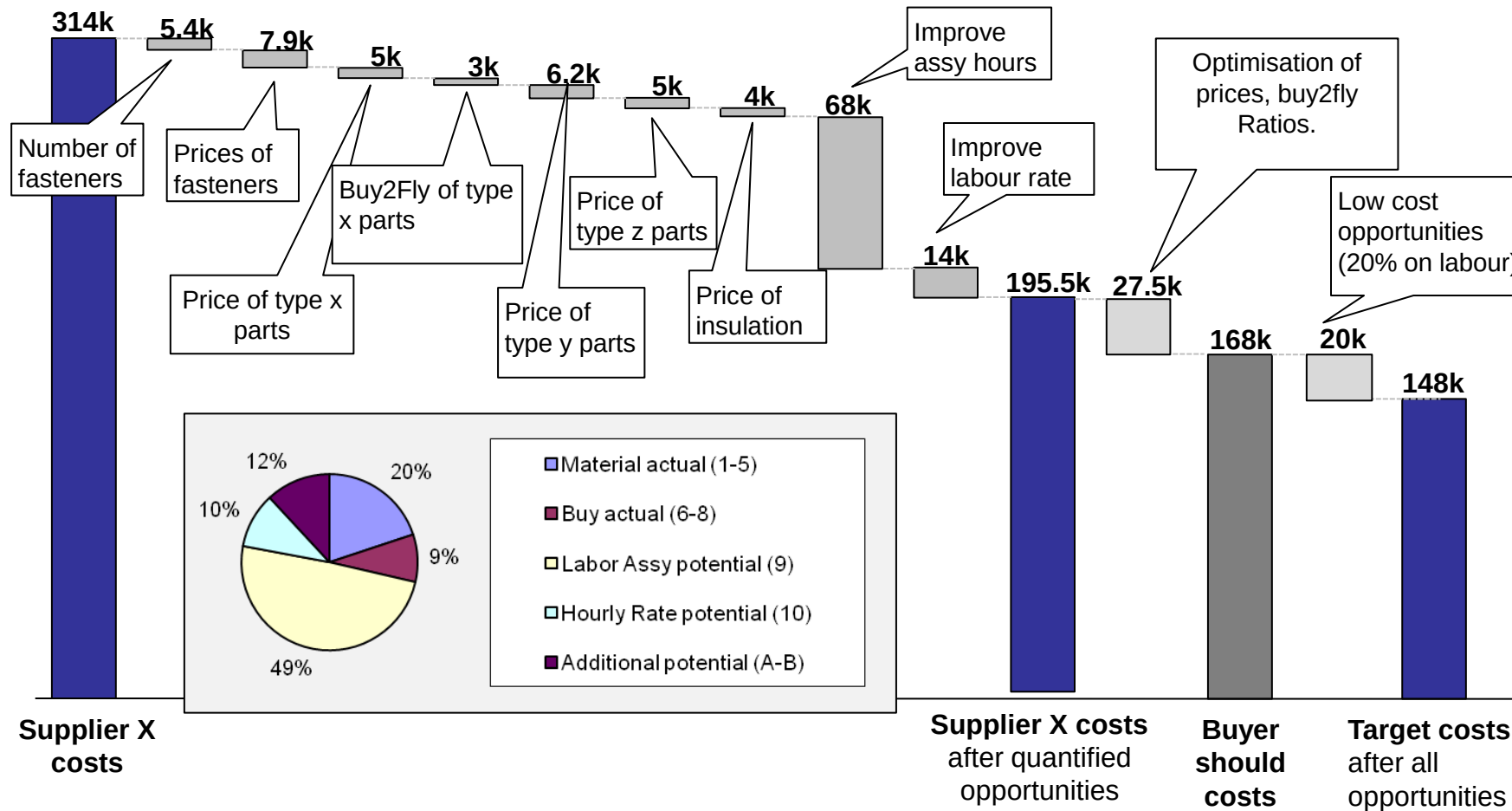


Assumptions

- 60 % in WE
- 40 % in Eastern Europe

Deep Dive gap analysis

Buyer & Seller



Questions to be answered during the Management meeting for BAFO preparation

Seller

— Definition

- Do we have the last technical definition?
- Do we have the last weight breakdown BAFO?
- How conservative are our numbers?
- Can we change material keeping the target weight and performance?
- Can we relax tolerances?
- Can we relax the ICY requirements, keeping the performance?
- Can we implement the outcome of the gap analysis?

— Industrial

- Can we find a better price for the materials?
- Do we have the state of the art technology?
- Are using the best manufacturing / assembly process for this WP?
- How conservative are our numbers?
- Can we think in a different industrial setup?
- Are our industrial means adequate?
- Can we implement the outcome of the gap analysis?

Questions to be answered during the Management meeting for BAFO preparation

— Financial / Contract

Seller

- Do we have enough financial capacity to undertake this task?
- Does our uplift / margin fits with the market?
- Can we adjust the price revision formula?
- Can we negotiate different payment scheme? Down payment?
- Does the contract type fits with our company strategy?
- Are our cost to price factors competitive?
- Any finding during the gap analysis?

— Program/Project

- Does this WP fits with the company development plans / strategy?
- Is our schedule feasible/credible ?
- Can we reduce the schedule?
- Do we have a solid risk analysis and mitigation plans defined?
- Is the negotiation margin adequate?

— Top Management final adjustment (Royal cut) and ...

BAFO RELEASE





ONLY the flight line: 15 Wide Bodies + 10 Single Aisle

Value of these aircraft: $15 \times 100 + 10 \times 50 = 2000 \text{ M€}$

Interest rate: 4%

PER MONTH cost: $(2000 \times 0,04)/12 = 6,7 \text{ M€}$

(fake values only for example)

1. Project Management Body of Knowledge 5th edition, Project Management Institute 2013.
2. Airbus internal documentation.
3. Roskam - Airplane-Design-(1985) Vol 8 - Airplane Cost Estimation Design, Development, Manufacturing